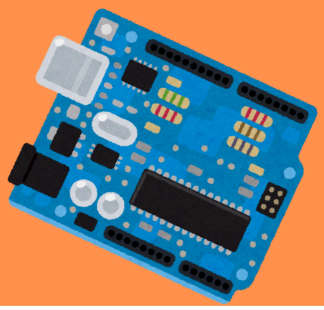
Topic 1

Topic Name: Explore Arduino Development Board



What is Arduino?

Small Ideas Big Possibilities!



Arduino is a small computer that helps you turn your ideas into real projects!

Arduino is an open-source electronics platform. It has a hardware board and easy-to-use software. You can write simple instructions (called code) and upload them to the board. Then, the board can control things like LED lights, motors, sensors and many other electronic components.

In simple words, Arduino helps you **think, code and create!**

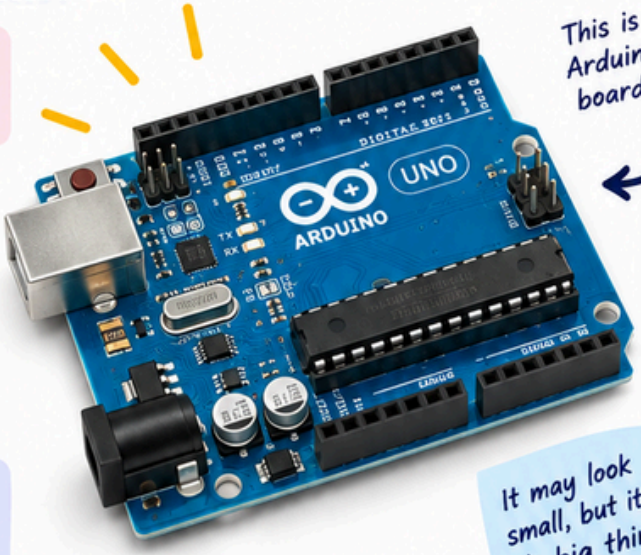
Why is Arduino Special?

Easy to Learn
Simple code and beginner-friendly.

Build Real Projects
You can make working models from your ideas.

Open Source
A global community shares ideas and helps you.

Fun and Creative
Use Arduino to solve real-life problems in unique ways.



This is an Arduino Uno board!

It may look small, but it can do big things!

What Can You Do with Arduino?



Make smart home models



Build robots and moving models



Create useful sensors



Design fun and creative projects

Ready to create something awesome?



The only limit is your imagination!



Arduino Uno Pin Configuration

Different Pins Big Possibilities!

The Arduino Uno has different pins to connect with sensors, motors, LEDs and many other electronic components. Each pin has a special job!

Reset Button
Resets the board and restarts your program.

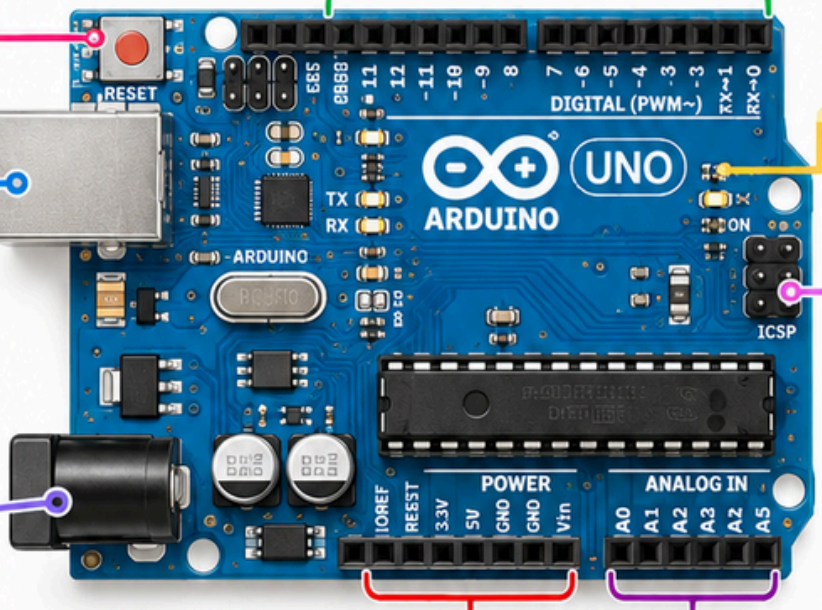
Digital Pins (0 – 13)
Used to connect digital devices like LEDs, buzzers, sensors etc.

USB Port
Used to upload code from your computer and also gives power to the board.

TX and RX Pins
Used for serial communication (with computer or other devices).

ICSP Header
Used for advanced programming (well-used with other boards).

Power Jack
Used to give external power (7–12V) to the board.



Power Pins
Provide power (5V, 3.3V), ground (GND) and other power-related connections.

Analog Pins (A0 – A5)
Used to read analog values from sensors like LDR, temperature sensor etc.

Each pin helps you connect, control and create!

Types of Pins on Arduino Uno

Digital Pins (0 – 13)



Used to give HIGH or LOW signals. Some of them (marked ~) can also be used for PWM (Pulse Width Modulation).

Analog Pins (A0 – A5)



Used to read analog values from sensors (values between 0 and 1023).

Power Pins



Include 5V, 3.3V, GND (Ground) and Vin. These pins provide power to the board and other components.

Special Pins



TX and RX for communication, Reset to restart the board and ICSP for advanced programming.





Explore the Features of Arduino Uno

Small Board Big Features! Endless Possibilities!

Arduino Uno is a powerful and easy-to-use microcontroller board. It comes with many useful features that help you connect sensors, LEDs, motors and other electronic components to build exciting projects.

Key Features of Arduino Uno

Microcontroller (ATmega328P)
The brain of the board. It runs your program.

14 Digital I/O Pins
Used to connect digital devices like LEDs, buzzers and sensors.

6 Analog Input Pins
Used to read analog values from sensors like LDR, temperature sensor, etc. (A0 – A5).

USB Connection
Used to upload code from your computer and also provides 5V power to the board.

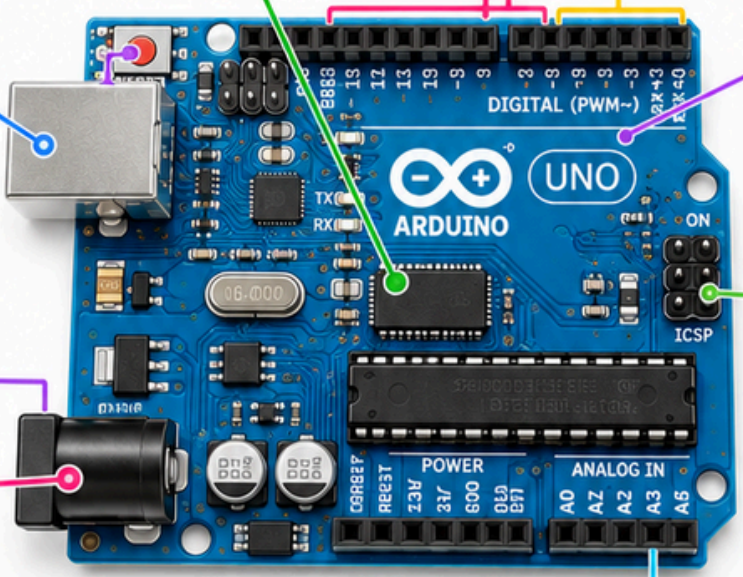
On-board LEDs
Shows the power, board status and activity.
Example: ON (power), L (built-in LED on pin 13).

Reset Button
Restarts the board and runs your program from the beginning.

ICSP Header
Used for advanced programming and updating the bootloader.

Power Jack
Used to give external power (7–12V) to the board.

Voltage Regulator
Keeps the power supply stable and provides the correct voltage to the board.



Why are these Features Important?

- Helps you easily connect different components.
- Makes it simple to upload and run your code.
- Provides both USB and external power options.
- Allows you to read signals from the real world using analog pins.
- Built-in features like LEDs and reset button make it user-friendly.

Fun Fact!

Arduino Uno is one of the most popular boards in the world because it is simple, affordable and perfect for beginners!



"Same board, endless ideas!"



Learning Outcomes

- The students will be able to explain in simple words what Arduino Uno is and how it helps us create electronic projects.
- The students will be able to identify the main pins on an Arduino Uno and tell what each type of pin is used for.
- The students will be able to recognise the important parts and features of the Arduino Uno, such as the USB port, power jack, LEDs, and microcontroller.
- The students will be able to understand how Arduino pins and features help us connect components like LEDs, sensors, and motors to build projects.